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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	Hands Off Our Rights
Selected Domain	Domain 1: Aiversecity Future-Ready Workforce
Selected SDG Track (1–17)	SDG 4 Quality Education; SDG 8 Decent Work and Economic Growth
Project / Solution Name	Future Skills
Name (Leader)	Hong Zhi Yang
Email & Phone (Leader)	zhiyang19015@gmail.com | +60 11-4035 9109
University Affiliation(s)	Universiti Tunku Abdul Rahman (UTAR)
Number of Team Members	3

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Malaysia's graduates often end up in jobs that do not use their education. In Q2 2026, 35.2% of employed people with tertiary education (1.95 million people) worked in semi-skilled or low-skilled jobs (DOSM, 2026b). Youth unemployment (ages 15–24) was 10.3% in 2024, against 3.2% nationally (DOSM, 2026a).

We focus on IT and computing undergraduates, who meet this problem when they choose a specialisation, an internship or a first job. "IT" is really six different areas: software engineering, data and AI, cybersecurity, cloud and networking, web and interface design, and IT management. Each needs different interests and skills, and AI is changing which roles are in demand faster than course guides are updated. Many students choose by hearsay.

The information they need exists but is scattered and not linked to the individual. TalentCorp's Malaysia Critical Occupations List (MyCOL) names occupations in short supply, DOSM publishes job-ad and wage data, and free courses are everywhere. No tool connects these to one student's own interests and skills, or turns a skills gap into a plan.

This matters economically and socially. Underemployment early in a career lowers lifetime earnings (DOSM, 2026a), Malaysia's critical IT occupations stay hard to fill, and students spend time and money on learning that leads nowhere.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

Future Skills is a web app that shows an IT student which IT area suits them, which of 26 IT roles fit, which skills they lack, and a free plan to learn them, with Malaysian demand for every role.

The student's journey:

1. **Profile:** an optional CV upload (or skill self-ratings), a 36-item interest questionnaire (RIASEC), then a skills test on only the 3–6 IT areas these point to (15–24 questions, including hands-on SQL).

2. **Area fit:** each area gets an interest and a skill score; results are grouped into "best fit", "can do" and "want to get there".
3. **Role matching:** 26 roles ranked by area fit, interest, skills and trend, with each score part and the missing skills shown.
4. **Reskilling pathway:** ordered free courses with an estimated finish time at the student's weekly study hours; HRD Corp-claimable paid courses listed separately.
5. **SDG and uncertainty:** every role and skill tagged to SDG targets; three future scenarios show how rankings could change.

How AI is applied:

- **GenAI CV reading:** an LLM (gpt-oss-120b) quotes CV evidence for each IT area; our code keeps only quotes really in the CV and scores them. The CV is never stored.
- **GenAI explanation:** "Explain this for me" turns one match's computed facts into plain language. A reply with any number not in those facts, or a prediction, is rejected and a rule-based summary shown. *GenAI explains; our data decides.*
- **Explainable matching and trend signals:** our scoring model combines O*NET skill data, MyCOL, DOSM job ads and arXiv AI-research counts, with a direction and confidence per signal.

Workshop skills we use: GenAI APIs (Fireworks AI, with output checking); data architecture (tables where every row carries a source and date, checked by a validator); cloud hosting (built and deployed on Replit, with an "Update data" job refreshing trend data).

3. Technical Architecture & Implementation Plan

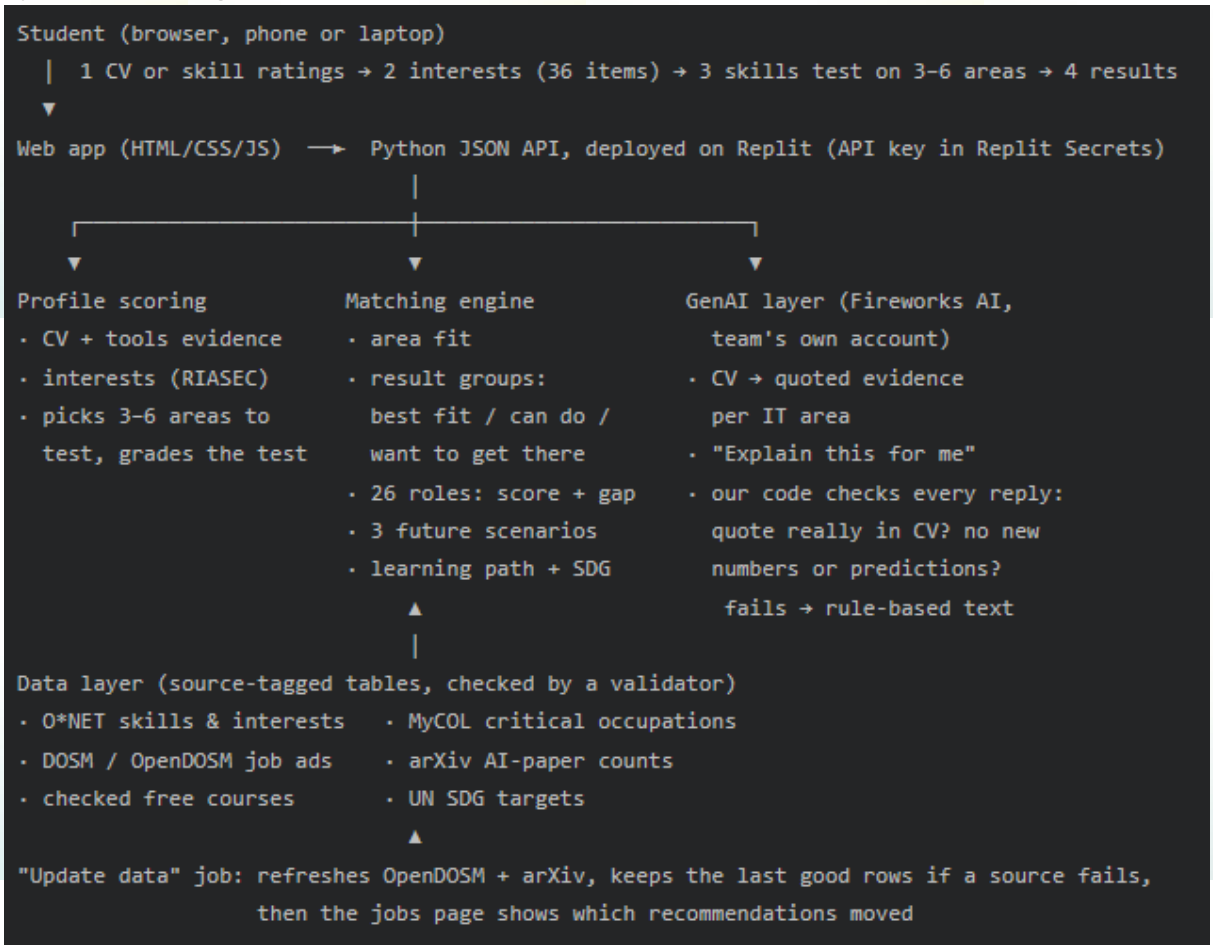
3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Layer	**Technology**
Frontend	HTML, CSS, plain JavaScript (no framework, no build step); works on phones
Backend	Python 3.12 standard library: a JSON API on http.server (no third-party packages)
Data	Structured CSV/JSON tables (roles, skills, role–skill links, courses, SDG map, trend inputs/signals/scores). Every row carries a source name, URL and date; validate_data.py checks them
GenAI API	Fireworks AI chat completions, model gpt-oss-120b (open-weight, Apache 2.0)
External data APIs	OpenDOSM API, arXiv API, Microsoft Learn Catalog API
Hosting / Dev	Replit (development, secrets, deployment). Live at https://future-skills.replit.app
Testing	Python unittest suite and a data validator

System Block Diagram



Sponsor GenAI API or Cloud Platform: Yes, Replit. We used Replit to develop the app, keep the API key in Replit Secrets and deploy the live prototype at https://future-skills.replit.app. Proof of use is attached separately. Our GenAI API is not sponsor-provided: gpt-oss-120b, an open-weight model, runs on our team's own paid Fireworks AI account. It reads students' CVs and explains each role match in plain language. If the key is missing, the app still works and shows rule-based text instead.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

Empathise: we are second-year IT students facing the same choice. Youth unemployment is 10.3% (DOSM), and the question we kept hearing was "where do my skills still matter?"

Define: a student needs roles they had not considered, proof that a trend is real, and a concrete next step.

Ideate: we chose transparent scoring over a black-box model so every recommendation can be checked, and set one rule: never invent a number.

Prototype: we designed the mobile flow on a clickable canvas first, then built and deployed it on Replit. A fictional demo student lets anyone try it in one click.

Test and iterate: students gave up on the full 24-question skills test, so we reordered the flow (CV → interests → test) and now test only the 3–6 areas they point to. 140 automated tests and a data validator run on every change.

Deployment by Phase 2: the app is already live, needs no packages, and still works without the GenAI key or when a data source fails. Next, we will pilot it with 20+ IT students, add free courses for uncovered skills, the next DOSM job-ads quarter, MyMahir occupation data, and tune the weights from pilot feedback.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	Target 4.4: "By 2030, substantially increase the number of youth and adults who have relevant skills, including technical and vocational skills, for employment, decent jobs and entrepreneurship." Supporting: Target 8.5, full and productive employment and decent work for all, including young people.
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	**35.2%** of employed people with tertiary education in Malaysia (1.95 million) work in semi-skilled or low-skilled jobs (skill-related underemployment), Q2 2026 (DOSM, 2026b). Youth unemployment is 10.3%, against 3.2% nationally (2024) (DOSM, 2026a). In our prototype, only **26 of the 186 skills (14%)** that our 26 IT roles need have a checked course in our catalogue (data/courses.csv, Oct 2026).
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	**1. Career clarity:** % of student testers who, after one session, can name a target IT role, its top missing skills and a next course to take, measured before and after the session. **2. Pathway coverage:** % of the 186 skills in our dataset with at least one manually checked free or public course.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation*	**1. Career clarity:** during Phase 2, at least 70% of 20 or more IT student testers can name a target IT role, its top

Metric Category	Your Response
e.g., "20% improvement in computational resource use."	missing skills and a next course after one session, compared with their before-session result. **2. Pathway coverage:** raise the share of the 186 skills with at least one manually checked free or public course from 14% (26 skills) to 40% (about 75 skills) by the Phase 2 submission.

Sources

- DOSM (2026a). *Labour Market Review, Q1 2026.* Department of Statistics Malaysia. https://storage.dosm.gov.my/analysis/lmr_2026-q1_en.pdf
- DOSM (2026b). *Labour Market Review, Q2 2026.* Department of Statistics Malaysia. https://storage.dosm.gov.my/analysis/lmr_2026-q2_en.pdf
- United Nations. *Global indicator framework for the Sustainable Development Goals and targets of the 2030 Agenda* (A/RES/71/313). https://unstats.un.org/sdgs/indicators/indicators-list/
- Future Skills team (2026). *Course catalogue, data/courses.csv* (checked October 2026). https://github.com/pekoremon/future-skills

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

☑ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.

☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.

☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.

☑ All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☑ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☑ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
O*NET 31.0 Database (US DOL/ETA)	Dataset	CC BY 4.0
DOSM Labour Market Review and Job Market Insights	Dataset	DOSM official release; attribution "Source: Department of Statistics Malaysia"
OpenDOSM (`productivity_qtr`, `wages_occupation`)	Dataset (API)	CC BY 4.0
TalentCorp MyCOL 2024/2025	Dataset (publication)	Public publication, cited
arXiv API (paper counts only)	Dataset (API)	Metadata CC0; arXiv API Terms of Use
Microsoft Learn Catalog API	Dataset (API)	Free to use; content © Microsoft
UN SDG Global Indicator Framework	Dataset	UN public document, cited

gpt-oss-120b (OpenAI), served by Fireworks AI	Pre-trained model	Apache 2.0 (model); Fireworks AI terms of service (API)
Free courses and videos (freeCodeCamp, CS50, OpenLearn, Linux Foundation, etc.)	Linked content (not copied)	© each provider; linked only
Python 3.12 standard library	Code	PSF License

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☑ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.



25/10/2026